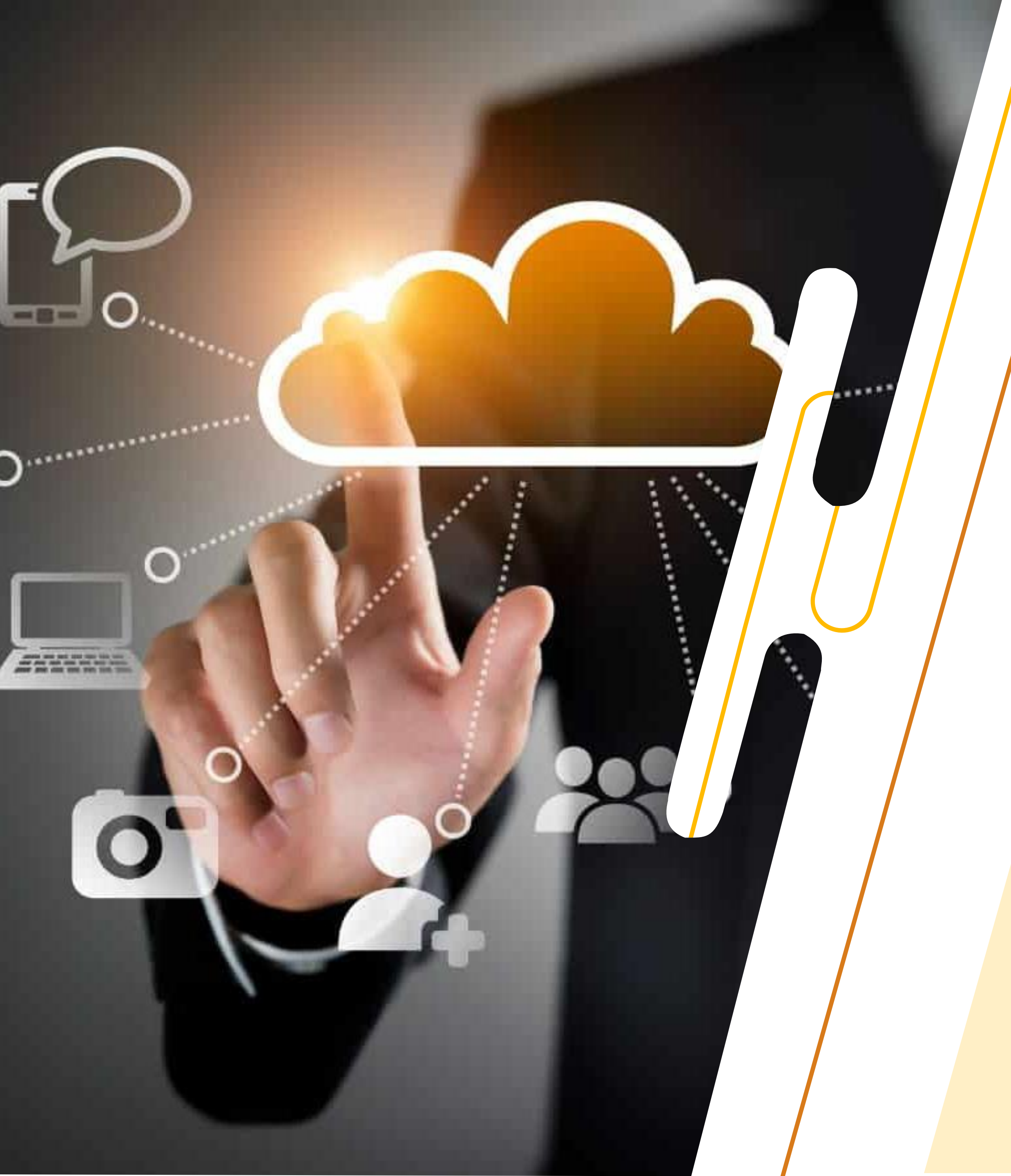


IT3061

Massive Data Processing and Cloud Computing

YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

Recap from last week

Exercise -

Find the

Reserved IP addresses = ?

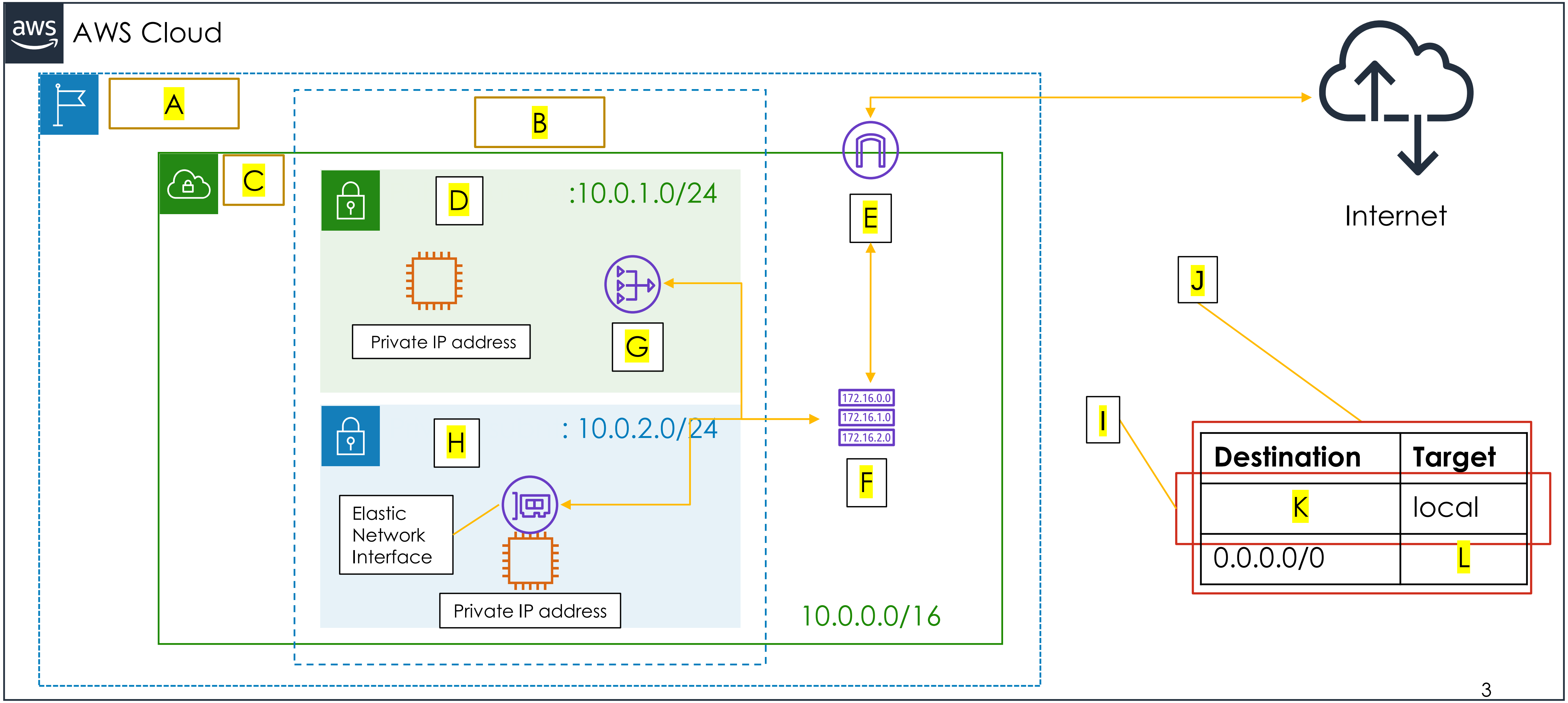
First usable IP address = ?

Last usable IP address = ?

Number of usable (Host) IP addresses = ?

of **172.16.10.128/25** and **192.168.0.64/27.**

Recap from last week



Lecture Outline

01. Cloud Storage

02. Benefits of Cloud Storage

03. Cloud Storage Services

04. Storage Types

05. Other aspects in Cloud Storage

Cloud Storage

- Cloud storage is a model of computer data storage in which the digital data is stored in logical pools
- The physical storage spans multiple servers or locations, and the physical environment is typically owned and managed by a hosting company and are responsible for keeping the data available and accessible, and the physical environment protected and running.

Benefits of Cloud Storage

- Store any amount of data and access it whenever you need from any device, without having to buy hardware and paying for storage that isn't be used.
- Storage can be provisioned rapidly, therefore any evolving requirements of the business can be addressed instantly, giving an organization a potential agility advantage.
- Can be used as storage for application resources, logs, media files, static website content, or any organizational data
- Variety of storage technologies to choose from for any organizational or application requirements
- Can be configured to support high availability, high I/O, redundancy, encryption and multiple geo-locations
- Data Management made easy

Cloud Storage Services

- AWS EBS
- AWS EFS
- AWS S3
- Azure Disks
- Azure Files
- Azure Blob Storage

Elastic Block Storage

Simple Storage Service (S3)

Elastic File System

Azure Blob Storage

Azure Files

Azure Disk Storage

Storage Types

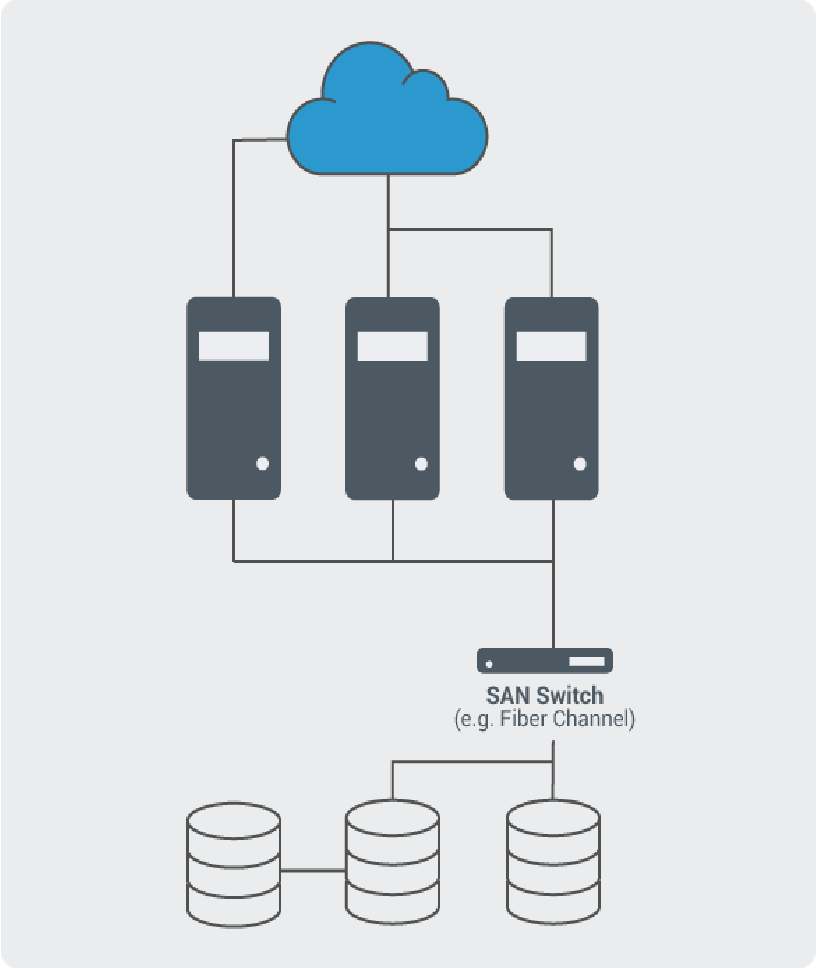
Block Storage

File Storage

Object Storage

Block Storage

- Block storage, sometimes referred to as block-level storage, is a technology that is used to store data files on Storage Area Networks (SANs) or cloud-based storage environments.
- Useful for computing situations where they require fast, efficient, and reliable data transportation.
- Breaks up data into blocks and then stores those blocks as separate pieces, each with a unique identifier.

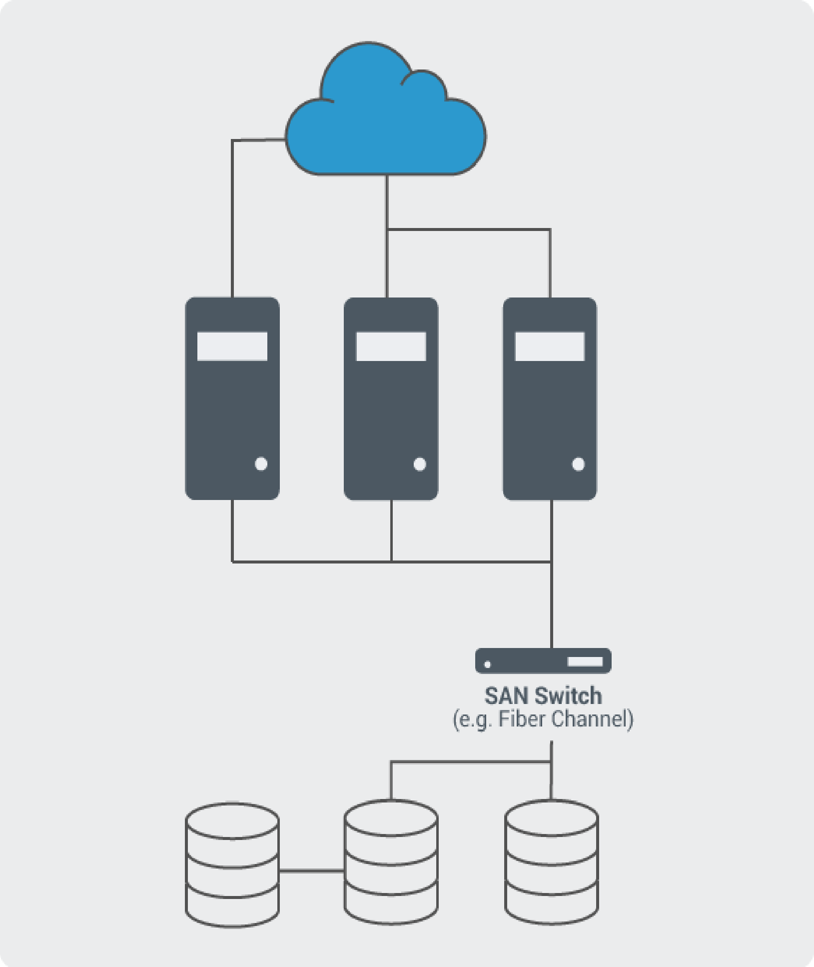


Block Storage

- The SAN places those blocks of data wherever it is most efficient.
- That means it can store those blocks across different systems and each block can be configured (or partitioned) to work with different operating systems.
- Examples –

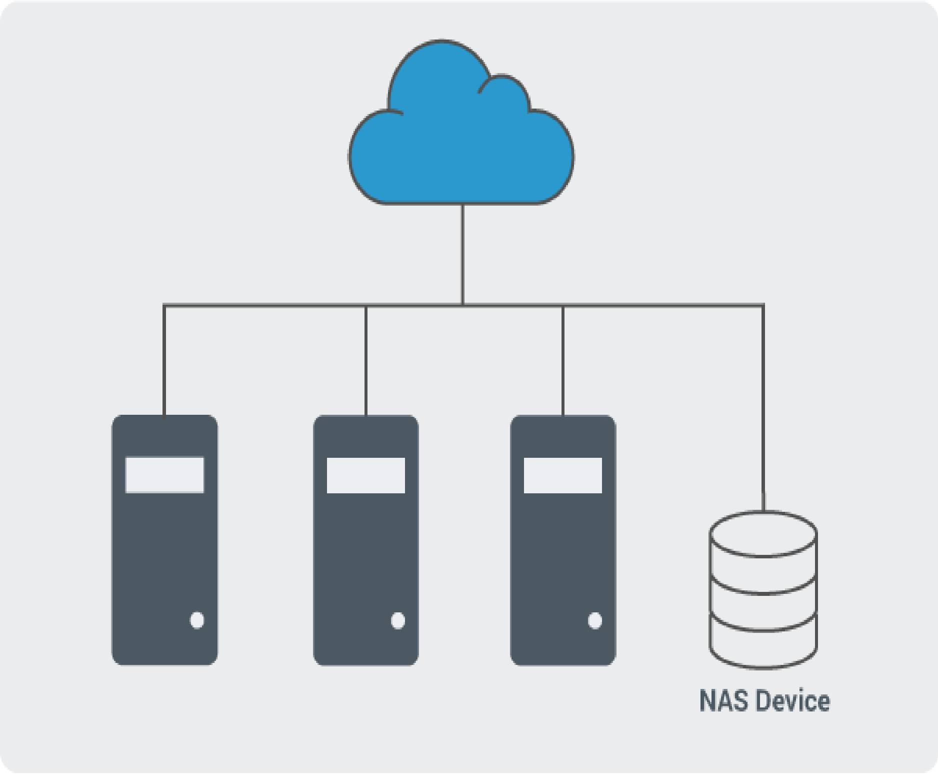
  AWS Elastic Block Storage (EBS)

  Azure Disks



File Storage

- File storage also called file-level or file-based storage is a hierarchical storage methodology used to organize and store data on a computer hard drive or on network-attached storage (NAS) device.
- In file storage, data is stored in files, the files are organized in folders, and the folders are organized under a hierarchy of directories and subdirectories.

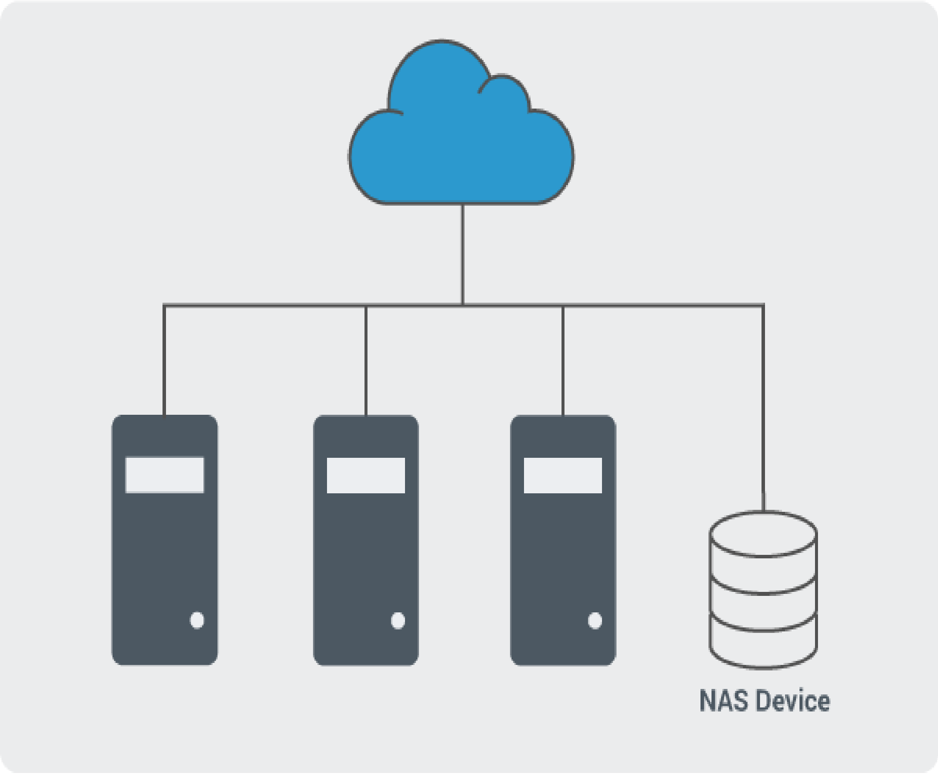


File Storage

- To locate a file, all you or your computer system need is the path from directory to subdirectory to folder to file.
- At its base, it is built with either SAS or SATA disks arranged in a RAID; it is then attached to devices over ethernet.
- Examples –

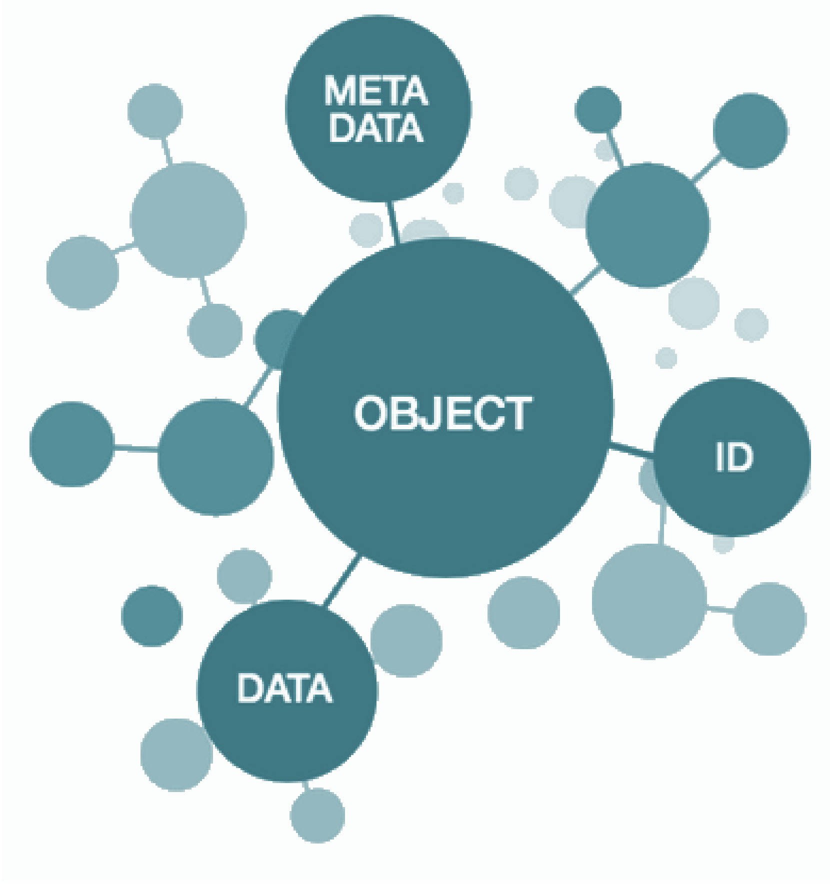
    AWS Elastic File System (EFS)

    Azure Files



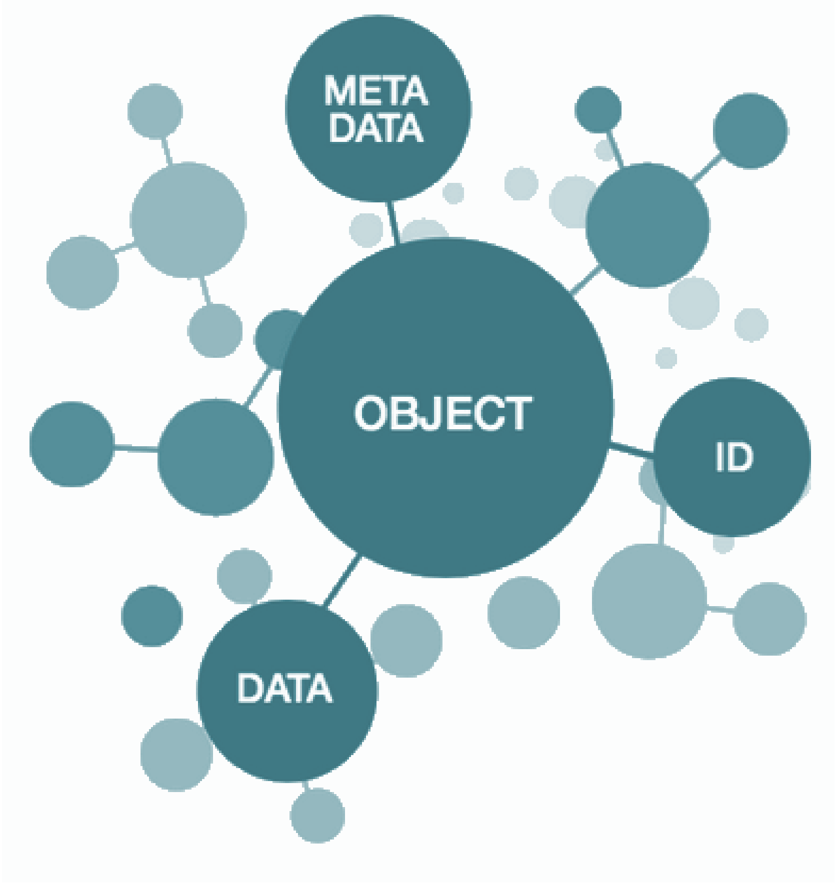
Object Storage

- Object storage is a computer data storage architecture that manages data as objects, as opposed to other storage architectures like file systems which manages data as a file hierarchy, and block storage which manages data as blocks within sectors and tracks.
- Each object typically includes
  - ✓ data
  - ✓ metadata
  - ✓ globally unique identifier



Object Storage

- Object storage systems allow retention of massive amounts of unstructured data.
- Object storage is used for purposes such as storing photos on Facebook, songs on Spotify, or files in online collaboration services, such as Dropbox.



Object Storage Tiers in AWS

- S3 Standard
- Infrequent Accessed (S3 – IA)

  For data that is accessed less frequently, but requires rapid access when needed. Cost-effective than S3, but charged a retrieval fee

- S3 One-Zone – IA

  Lower-cost option for infrequently accessed data, but do not require multiple AZ data resilience

Object Storage Tiers in AWS

- S3 – Intelligent Tiering

  Designed to optimize costs by automatically moving data to the most cost-effective access tier, without performance impact or operational overhead

- S3 Glacier

  Low-cost class for data archival. Retrieval may take minutes to hours. ‘Glacier-Deep Archive’ s more cost efficient, but may take more than 12 hours for data retrieval

Object Storage Tiers in Azure

- Hot tier

  An online tier optimized for storing data that is accessed or modified frequently. The hot tier has the highest storage costs, but the lowest access costs.

- Cool tier

  An online tier optimized for storing data that is infrequently accessed or modified. Data in the cool tier should be stored for a minimum of **30** days. The cool tier has lower storage costs and higher access costs compared to the hot tier.

Object Storage Tiers in Azure

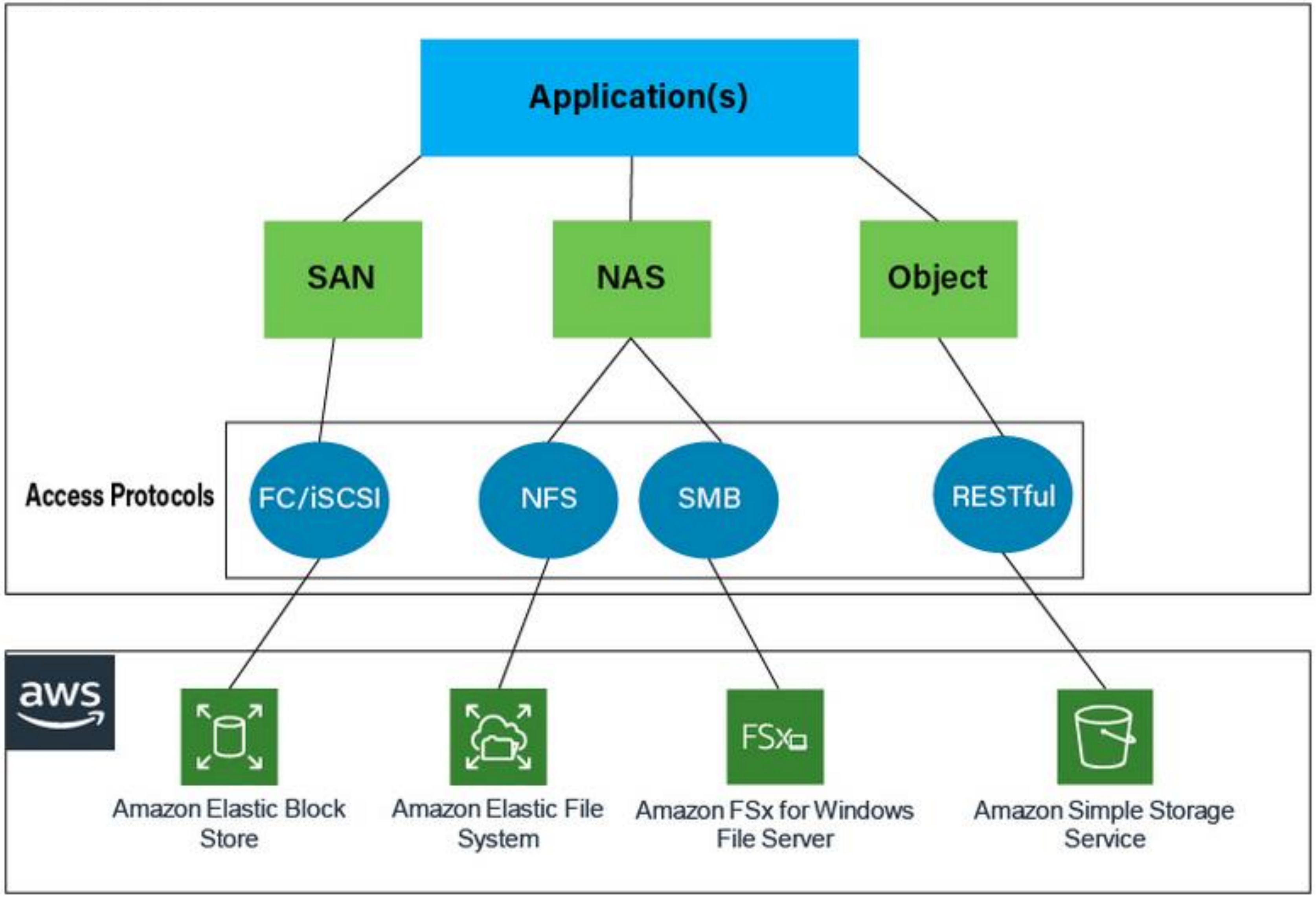
- Cold tier

An online tier optimized for storing data that is rarely accessed or modified, but still requires fast retrieval. Data in the cold tier should be stored for a minimum of **90** days.

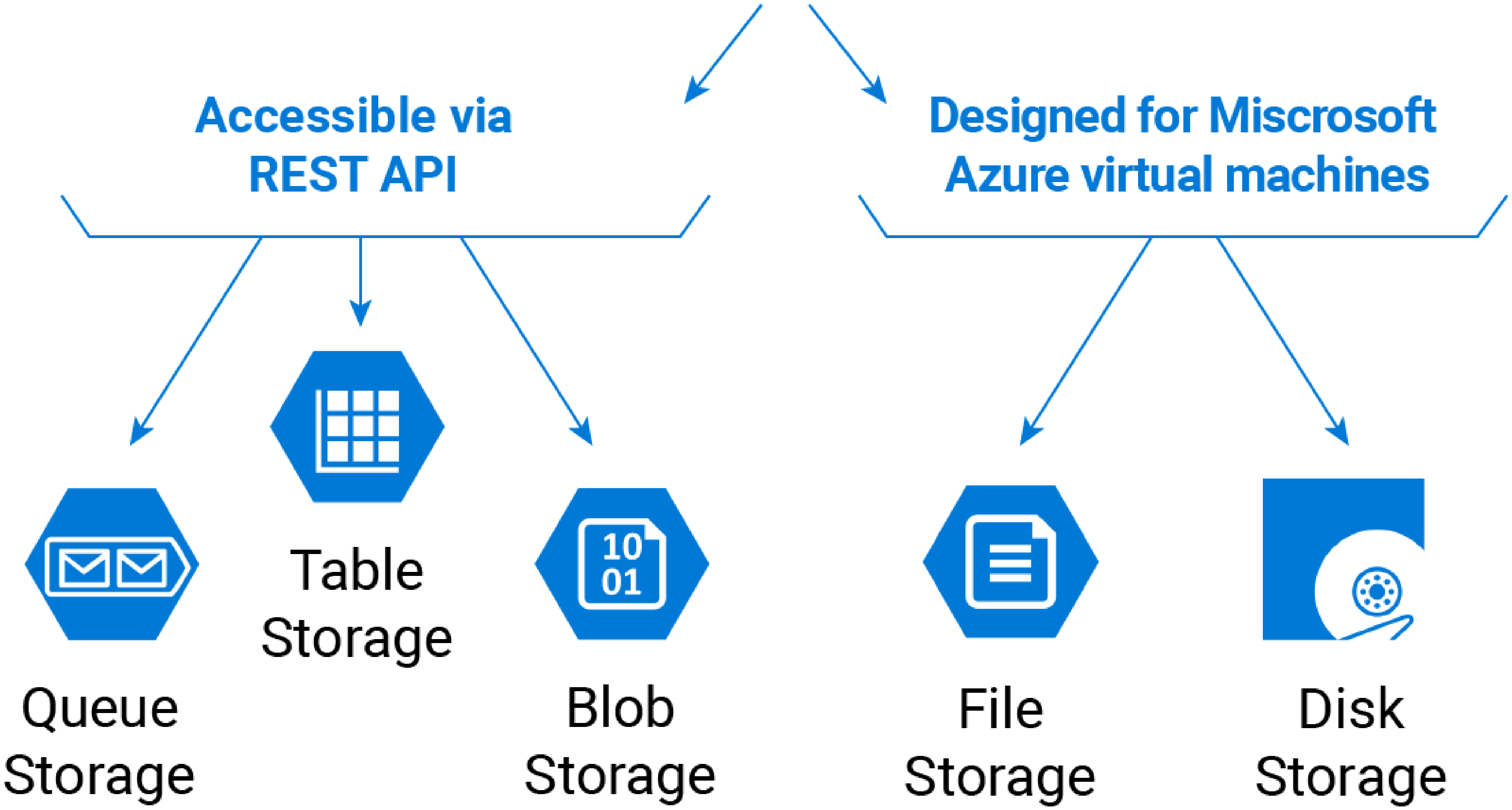
- Archive tier

An offline tier optimized for storing data that is rarely accessed, and that has flexible latency requirements, on the order of hours. Data in the archive tier should be stored for a minimum of **180** days.

AWS Storage Options



Azure Storage Options



Other aspects in Cloud Storage

Archiving and Backups

Hybrid Storage

Bulk Data Transfer

Other aspects in Cloud Storage

Example – Data Transfer service in AWS

AWS Snowball

- Petabyte-scale data migration
- Rugged 8.5 G impact case
- Rain and dust resistant
- Data encryption end-to-end
- 80 TB capacity/10 Gb network
- Move 250 TB in 1 Week!

AWS Snowball Edge

- Compute and storage for hybrid/edge workloads
- 100 TB local storage
- Local compute equivalent to an Amazon EC2 m4.4xlarge instance
- 10GBase-T, 10/25 Gb SFP28, and 40 Gb QSFP+ copper, and optical networking
- Ruggedized and rack-mountable

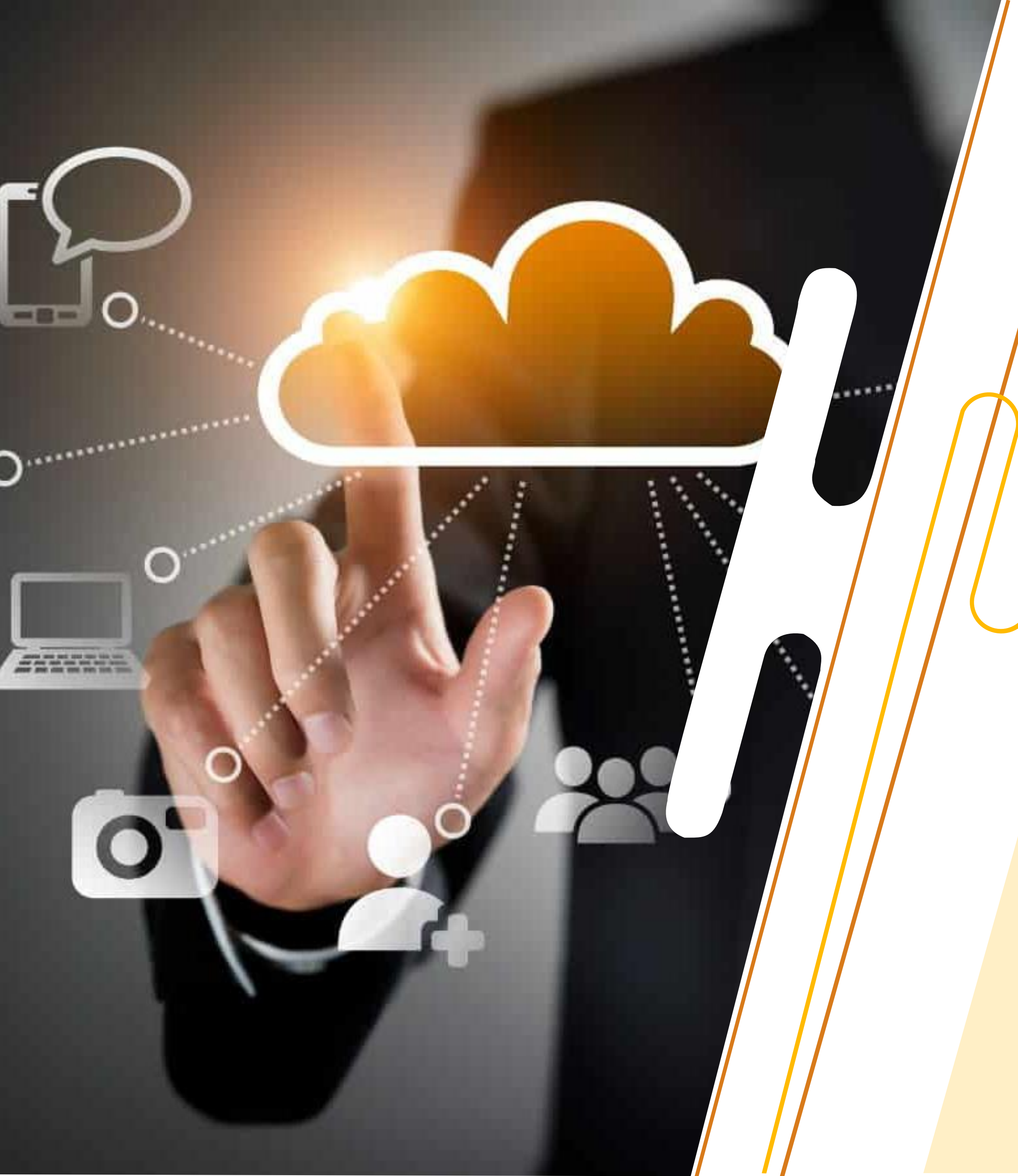
AWS Snowmobile

- Exabyte-scale data migration
- Up to 100 PB capacity
- Data encryption end-to-end
- Dedicated security personnel
- GPS tracking, alarm monitoring, 24/7 surveillance, and optional additional security

Demo

References

- AWS Academy Cloud Foundations
- AWS Documentation
- Azure Documentation



Next Session : Load Balancing and Auto Scaling